
Time-Dependent Perturbation Theory

- We attempt to obtain:

$$|\psi(t)\rangle = \sum_n c_n(t) e^{-iE_n^{(0)}t/\hbar}$$

through

$$i\hbar \frac{dc_m(t)}{dt} = \sum_n c_n(t) e^{i(E_m^{(0)} - E_n^{(0)})t/\hbar} \langle \phi_m | \lambda H^{(1)}(t) | \phi_n \rangle$$

with the assumption $c_n(t) \rightarrow \delta_{nk}$ as $t \rightarrow 0$,

$$c_m(t) = \frac{1}{i\hbar} \int_0^t dt e^{i(E_m^{(0)} - E_k^{(0)})t/\hbar} \langle \phi_m | \lambda H^{(1)}(t) | \phi_k \rangle$$

- *Homework 1 Problem 3*
 - *Homework 1 Problem 4*
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The Interaction of Charged Particles with the Electromagnetic Field

- The Hamiltonian is upgraded:

$$H = \frac{\mathbf{p}^2}{2m} \mapsto H = \frac{1}{2m} (\mathbf{p} + q\mathbf{A})^2 + qV$$

(or $q\phi$)

- The Berry connection of a eigenstate is:

$$A_\mu^{(n)} = i \langle n | \partial_\mu | n \rangle$$

while the Berry curvature of a eigenstate is:

$$F_{\mu\nu}^{(n)} = \partial_\mu A_\nu^{(n)} - \partial_\nu A_\mu^{(n)}$$